

xai_adapter: Architecture & Design Patterns

Class Hierarchy

XAIAdapter (Abstract)
fit() | explain() | attribute()

Attribution

SurrogateMethod

Design Patterns

1

Template Method
Base class defines skeleton; subclasses fill in steps

2

Strategy
Switch between Attribution vs Surrogate seamlessly

3

Registry

✓ Extensible (add new methods without touching base) ✓ Unified Interface (all methods return XAIAdapterResult) ✓ Pluggable I/O (custom algorithms via make_attribution())

Extension Points & Data Flexibility

Three Extension Paths

1. Subclass Attribution

Implement custom `explain()` method

When: New local attribution algorithm

2. Subclass SurrogateMethod

Implement custom `fit()` & `explain()`

When: New rule-based or surrogate

3. Register Custom Callable

Wrap via `make_attribution(fn)`

When: Quick integration of existing code

Input Flexibility

Raw Data (X, y) → Train on fly → Attribution & Surrogate methods

Precomputed Tables (explanation_df + metadata_df) → Surrogate methods

✓ Both Paths → `XAIAdapterResult { values (n_instances × n_features) | base_values | method | metadata }`